

1.

Find eigenvalues and eigenvectors of the matrix

$$\begin{bmatrix} 3 & -2 & -1 \\ 0 & 5 & 1 \\ 0 & 2 & 4 \end{bmatrix}$$

2. Write the characteristic polynomial and the eigenvalues of the matrix:

$$A = \begin{bmatrix} 6 & 0 & 0 & 0 & 0 \\ 9 & -4 & 0 & 0 & 0 \\ -2 & 0 & 11 & 0 & 0 \\ 1 & -1 & 3 & 0 & 0 \\ 0 & 1 & -7 & 4 & 8 \end{bmatrix}$$

3. Find the characteristic polynomial, eigenvalues and eigenvectors of

$$A = \begin{bmatrix} -4 & 2 \\ 3 & -5 \end{bmatrix}$$

4. Let A be the matrix of a linear transformation, which rotates all vectors in $\mathbb{R}^2$ counterclockwise through an angle of $\pi/6$. Explain, using the geometry of eigenvectors, why A cannot have any real eigenvalues.

5. Given matrix $A = \begin{bmatrix} 3 & -2 & -1 \\ 0 & 5 & 1 \\ 0 & 2 & 4 \end{bmatrix}$, find matrices P and D such that D is a diagonal matrix and $A = PDP^{-1}$